Carbon and Isotope Transport in CO_2 Electrolysis

Hanyi Seok

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Objective and contribution

I developed coupled carbon and isotope balances to interpret outlet ^{13}C enrichment in CO_2 electrolysis and distinguish absorption, reaction, membrane transfer, and liquid storage. For an anion-exchange-membrane (AEM) gas-diffusion-electrode (GDE) reactor, my work combined model formulation, archived-data analysis, measurement design, and parameter-estimation workflows. The model links well-mixed liquid inventories to quasi-steady gas observations.

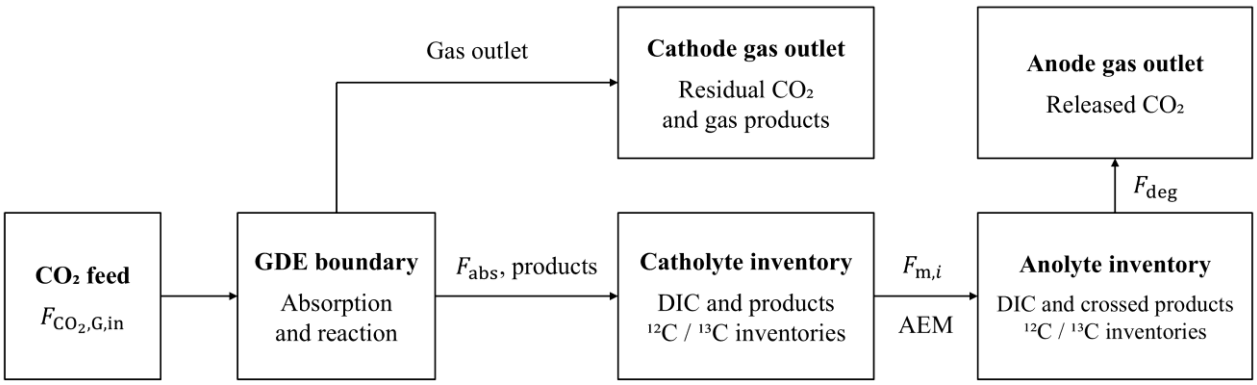


Figure 1. Carbon and isotope pathways. DIC denotes dissolved inorganic carbon.

Carbon and isotope balances

DIC comprises dissolved CO_2, bicarbonate, and carbonate; products are tracked separately. Its catholyte balance couples absorption, reaction, membrane transfer, and external flows:

$$\frac{\mathrm{d}N_{\mathrm{DIC,C}}}{\mathrm{d}t} = F_{\mathrm{abs}} - \sum_j a_j f_j r_j - F_{\mathrm{m,DIC}} + F^{\mathrm{ext}}_{\mathrm{DIC,C}}$$

$N_{\mathrm{DIC,C}}$ is the DIC-carbon inventory; F_{abs} and $F_{\mathrm{m,DIC}}$ are absorption and membrane-carbon rates. Product j forms at molecular rate r_j, contains a_j carbon atoms, and draws carbon fraction f_j from DIC; the remainder is gas-fed. The external term is net input minus output, including sampling. Summing compartment balances cancels internal crossover, leaving the reactor-wide carbon account.

Each carbon pool also carries a ^{13}C inventory, allowing isotope-specific fluxes to change composition even when their total-carbon contributions cancel. Product labels combine source fractions and assumed reaction isotope rate ratios. For DIC-derived products, the precursor label comes from isotope-resolved dissolved CO_2 rather than bulk DIC. Independent source and product labels are needed to separate mixing from reaction discrimination.

Under anion-dominated charge transport, membrane transfer combines open-circuit and current-driven contributions:

$$F_{\mathrm{m},i} = F^0_{\mathrm{m},i} + \frac{I_{\mathrm{F}}}{F}\frac{t_i}{|z_i|}$$

$F^0_{\mathrm{m},i}$ is the open-circuit net transfer rate; I_{F} is Faradaic current, F the Faraday constant, $|z_i|$ the charge magnitude, and t_i the species' fraction of anionic current. These fractions sum to one and depend on composition and effective carrier weights. Neutral species retain passive transfer only. Paired liquid inventories and matched passive/current contrasts constrain net transfer or selected coefficient combinations.

Analysis of archived observations

I fitted a reduced gas–liquid absorption model to 27 archived zero-current bubbling conditions (median pH RMSE 0.180). Carbonate equilibrium at fixed nominal alkalinity converts modeled DIC to pH. The selected fit has RMSE 0.050; its residuals assess agreement within the retained window in a separate bubbling vessel.

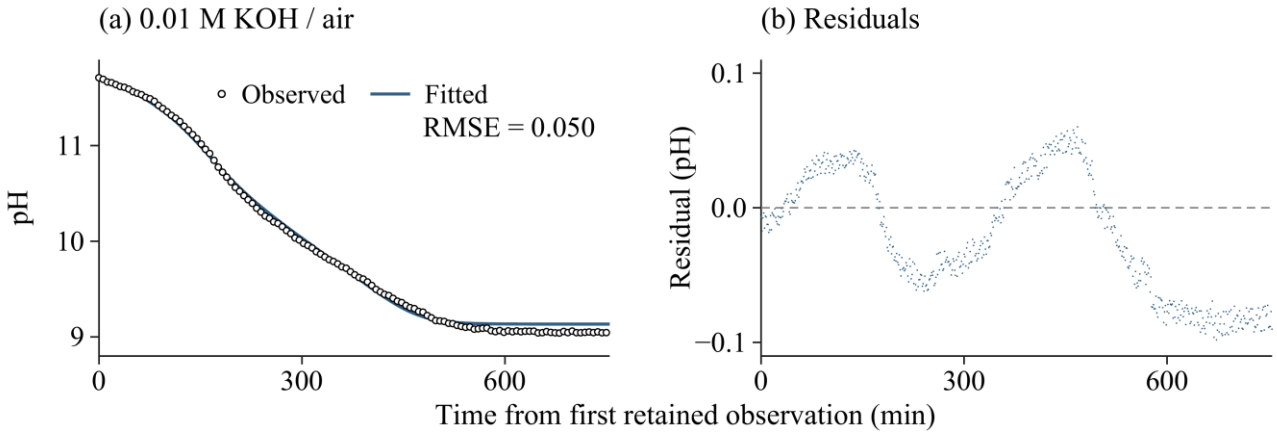


Figure 2. Reduced-model fit to archived bubbling data. (a) Observed and fitted pH for 0.01 M KOH bubbled with air (RMSE = 0.050). (b) Residuals, observed minus fitted pH. This within-sample fit uses a manually selected observation window in a separate bubbling vessel.

Analysis of 28 digitized anode-gas observations showed strong assumption sensitivity: reducing assumed oxygen Faradaic efficiency from 1.00 to 0.95 shifted the effective bicarbonate–carbonate factor from 5.843 to 9.312, with nearly identical carbon-per-charge ratio RMSE (0.0188 versus 0.0186). Similar fits therefore did not uniquely determine the transport mechanism.

For archived outlet-isotope data, re-normalizing the same current-on observations against two 75 sccm baselines changed the inferred ^{12}C and ^{13}C response slopes. This motivated matched baselines, calibrated gas rates, and independent source labels. These separate comparisons assess reduced models rather than validate the full reactor framework.

Measurement design and computational results

I proposed calibrated gas rates and direct DIC for absorption, since pH-derived DIC is not independent. O_2 and integrated charge would test the efficiency assumption; paired liquid inventories and anode release would separate net membrane delivery from storage, degassing, and possible product oxidation. Source and product isotope measurements would distinguish carbon-source mixing from reaction discrimination.

I implemented Python/Jupyter workflows with eight targeted effective-parameter estimators, evaluated through synthetic recovery, held-out prediction, and local rank and uncertainty checks. With source labels and reaction isotope ratios fixed, a prescribed synthetic DIC-source fraction of 0.350 was estimated as 0.3498 ± 0.0028 (conditional standard error). Large carrier-weight uncertainties revealed weak practical separation despite full local rank. No new experiments were conducted; intrinsic coefficients and full-reactor validation require independent measurements.